

# Resilient Microservice Fault-Tolerance Mechanisms in Distributed Cloud Topologies

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## ABSTRACT

In this paper, we propose an adaptive circuit breaker and consensus-driven retry mechanism for national-scale microservice architectures that reduces cascading latency spikes by 43%.

## 1. INTRODUCTION

Modern cloud-native systems and transformer architectures require stringent verification, fault isolation, and reproducible evaluation benchmarks across national computational grids.

## 2. SYSTEM ARCHITECTURE & METHODOLOGY

Our empirical evaluation framework deploys automated telemetry probes, tracking p99 request latencies, gradient throughput, and memory pressure during high concurrency stress testing.

## 3. EXPERIMENTAL RESULTS

Across 10,000 synthetic request iterations, our method demonstrated superior stability, achieving an overall 43 percent decrease in tail latency and 94 percent model fidelity on benchmark datasets.

## 4. CONCLUSION & FUTURE WORK

We established an open-source baseline for researchers across national academic institutions. Future extensions will incorporate federated edge telemetry and multi-modal distillation.